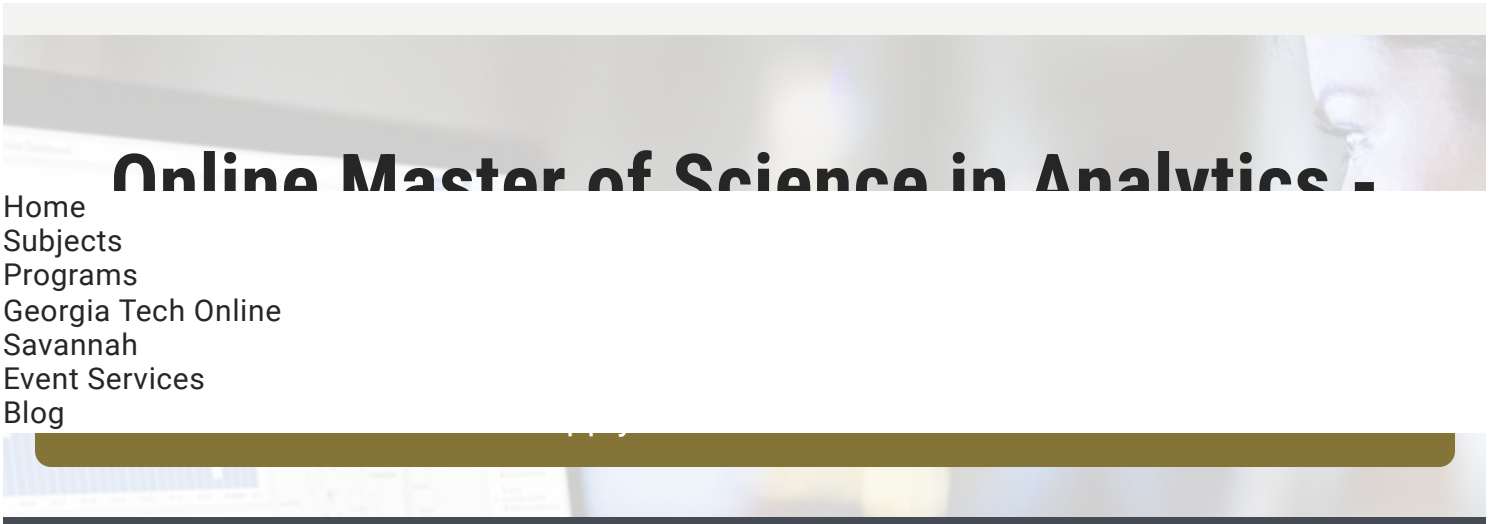
Georgia Tech

Professional Education

College of Lifetime Learning



Curriculum

There's no better way to learn about a field that requires computing, business, statistics, and research, than through a university that ranks in the top 10 in all four categories. The Online Master of Science in Analytics (OMS Analytics) at Georgia Tech meets this criterion – and many other high standards.

Many students fulfill the requirements for this online data analytics master's degree in one-and-a-half to two years; however, the program is flexible enough that you have up to six years to complete it.

The program also consists of 30 course offerings. All OMS Analytics students take 15 hours of mandatory (core) coursework on big data analytics in business, visual analytics, computing, statistics, and operations research essentials. Additionally, you get to focus on your individual interests through 15 hours of electives in a specialized track (area of concentration).

Six-Hour Practicum

The Online Master of Science in Analytics program culminates with a practicum course that enables you to apply previously learned concepts and classroom teachings to a project of significant interest to you. You can propose an authentic business need that uses data from your own employer or third party (external project), or you can gain real-world experience through one of our GT-sponsored projects (internal project).
The practicum is 10-15 weeks depending on the semester and the process starts several months in advance.

- **Fall** (15 weeks): Process begins in **mid-May** with the *Practicum Interest Survey* sent to the student
- **Spring** (15 weeks): Process begins in **mid-September** with the *Practicum Interest Survey* sent to the student
- **Summer** (10 weeks): Process begins in **mid-February** with the *Practicum Interest Survey* sent to the student

Visual Snapshot

Our OMS Analytics curriculum grid breaks down the different types of courses and concentrations into digestible components. For more details about each track's focus, scroll down to the section below.

Areas of Specialization

Every student in the degree program pursues an area of specialization within data science and analytics. Three focus areas are available: an Analytical Tools track, a Business Analytics track, and a Computational Data Analytics track. The summaries below describe each in more detail.

FALL 2027 PRIORITY REVIEW APPLICATION DEADLINE

Feb 1, 2027

FALL 2027 FINAL APPLICATION DEADLINE

Mar 15, 2027

FALL 2026 PROGRAM START

Aug 24, 2026

NEXT INFO SESSION

Nov 4, 2026

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Analytical Tools Track

The Analytical Tools track focuses on the quantitative methodology: how to select, build, solve and analyze models using methodology, regression, forecasting, data mining, machine learning, optimization, stochastics, and simulation.

Introductory Core Requirements (9 hours)

Special Note

Students with sufficient background in any of these areas may apply to replace one or more of these courses with an elective.

Introduction for Computing for Data Analytics (CSE 6040)
Computational techniques needed for data analysis; programming, accessing databases, multidimensional arrays, basic numerical computing, and visualization; hands-on applications and case studies. Credit will not be awarded for both CSE 6040 and CX 4240.

Introduction to Analytics Modeling (ISYE 6501)
This course gives a basic introduction to a wide variety of analytics models and techniques, including the basic ideas behind the models, experience using software to solve/analyze them, and case studies dealing with combining models to find a complete solution. Modeling approaches covered include classification, clustering, change detection, time series modeling, regression models, design of experiments, probability distributions, probability-based models and simulation, PCA, and optimization. Cross-cutting topics like data preparation, model validation, and variable selection are also covered.

Business Fundamentals for Analytics (MGT 6201)
An accelerated introduction to the basics of management and the language of business. The course provides a framework that will enhance a person's effectiveness in the business world.

Advanced Core Requirements (6 hours)

Statistics Electives (6 hours)

Operations Elective (3 hours)

Track Electives (6 hours)

Additional Electives (0-9 hours)

Applied Analytics Practicum (6 hours)

Business Analytics Track

The Business Analytics track explores the understanding, framing, and solution of problems in marketing, operations, finance, management of information technology, human resources, and accounting in order to develop and execute analytics projects within businesses.

Introductory Core Requirements (9 hours)

Special Note

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Operations Research Electives (3 hours)

Track Electives (6 hours)

Additional Electives (0-9 hours)

Applied Analytics Practicum (6 hours)

Computational Data Analytics Track

The Computational Data Analytics track explores a deeper understanding of big data—including how to acquire, preprocess, store, manage, analyze, and visualize large datasets.

Introductory Core Requirements (9 hours)

Special Note

Students with sufficient background in any of these areas may apply to replace one or more of these courses with an elective.

Introduction for Computing for Data Analytics (CSE 6040)
Computational techniques needed for data analysis; programming, accessing databases, multidimensional arrays, basic numerical computing, and visualization; hands-on applications and case studies. Credit will not be awarded for both CSE 6040 and CX 4240.

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Stay up-to-date on important deadlines, updates, and news. Also get tips for writing a stellar application!

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Remote learning doesn't mean missing out. Be part of our growing social network with other students.

Questions?

Want to know how an OMS Analytics degree can help you advance your career? Want to stay current on important deadlines, updates, and news? Need more details?

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Georgia Institute of Technology

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Professional Education

Georgia Institute of Technology

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Atlanta, GA 30308

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